

# ANON

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## ABSTRACT

LLM agent frameworks increasingly expose model invocations through structured programming abstractions, with framework runtimes lowering program-level constructs into concrete prompts. We observe that these programmatically generated prompts are inherently heterogeneous: their components exhibit different sharing scopes, lifetimes, and evolution patterns across invocations. However, existing serving systems treat prompts as opaque text, losing this structure and missing opportunities for more effective KV-cache reuse and management. To address the gap, we present ANON, a compiler-serving co-design that recovers and exploits this heterogeneity for efficient KV-cache management.

## 1 INTRODUCTION

LLM agents invoke a series of dependent model calls with share context. Reusing the key-value (KV) cache across these calls avoids redundant prefill computation, making prompt reuse an important opportunity for efficient agent serving. Existing agent serving systems primarily optimize at the level of requests or LLM invocation sequences. Pythia (Yu et al., 2026) profiles agent workflows to guide cache management, request scheduling, and autoscaling. CacheScout (Zhang et al., 2026) learns execution transitions online to guide eviction and prefetching, while PBKV (Zheng et al., 2026) predicts future invocations and estimates the reuse potential of cached requests. These systems exploit predictability between invocations, but treat the contents of each prompt as an opaque token sequence.

However, we observed that prompts in agent LLM invocations are not monolithic, they differ in three dimensions: (1) Sharing scope: one value can be shared across multiple invocations, or only within a single invocation. (2) Lifetime: Different bindings have different lifetimes. Some bindings are single-shot volatile inputs, while others may be reuse across the whole session. (3) Evolution pattern: context for LLM calls can evolve differently overtime, such as stay fixed, being replaced, or accumulating across invocations. We name this as **intra-prompt heterogeneity**.

Neglecting this intra-prompt heterogeneity can lead to sub-optimal KV-cache management. Consider a financial analysis agent that answers a question about a company’s filings

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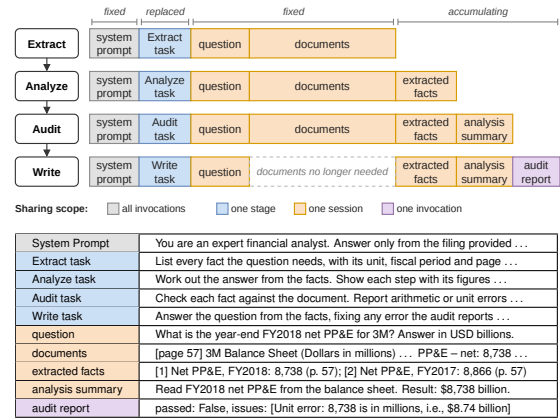


Figure 1. An example of intra-prompt heterogeneity in a financial analysis agent.

in four stages: Extract, Analyze, Audit, and Write (Figure 1). The prompt of each stage consists of a system prompt, a task description, the question, the documents, and the outputs of earlier stages. The system prompt is shared by every invocation, whereas the task description is replaced at every stage. The question and the documents remain fixed, but only the first three stages read the documents. The outputs of earlier stages accumulate as the session proceeds. Mainstream KV-cache serving engines are prefix-based: reuse stops at the first token that differs. Because the task description precedes everything else, these stages only share system prompt, and Analyze and Audit recompute the KV states of the documents although their contents remain unchanged. Moreover, without knowing which components persist or evolve, the serving system cannot accurately determine which cached tokens should be retained, evicted, or prefetched. In this example, it cannot tell that the documents can be released after Audit. Neither problem is easy to solve in the serving

system alone, because the semantic boundaries and reuse patterns are hard to identify from token sequences.

This information, however, is available when the prompt is programmatically generated. Agent frameworks increasingly expose LLM invocations through structured programming abstractions, such as byLLM (Dantamarayana et al., 2025), DSPy (Khatab et al., 2024), and NOOA (Furgale et al., 2026). These frameworks let developers declare each LLM invocation as a function with typed parameters, and the framework runtime generates the prompt from this declaration and the argument values. The sharing scope, lifetime, and evolution pattern of each component are therefore largely determined by the program, which can be systematically analyzed. This provides an opportunity to identify and exploit intra-prompt heterogeneity.

We present ANON, a compiler-serving co-design that recovers intra-prompt heterogeneity from the agent program and exploits it for efficient KV-cache management. The compiler analyzes the program and passes the recovered heterogeneity to the serving system, which addresses both problems above: it determines prompt layouts that expose additional prefix reuse, and keeps the resulting KV states resident when they are likely to be needed. Identifying and exploiting intra-prompt heterogeneity introduces two challenges:

**First, heterogeneity is not always explicit in prompt.** The sharing scope and lifetime of bindings are closely bonded with the program-level structure and actual runtime assignment. Determining which layout best serves prefix caching is therefore not trivial. ANON combines static analysis with runtime information to determine prompt layouts. The compiler derives a partial order over prompt components based on their sharing scopes and lifetimes. Then, ANON uses the actual token lengths of bindings observed from log to order components left unconstrained by this partial order, producing a concrete layout that improves prefix cache efficiency.

**Second, exploiting future reuse requires coordinating cache retention and prefetching.** Prompt components differ in when they will be reused and how much recomputation their cached prefixes can avoid. Retaining these prefixes consumes capacity needed by other requests, while prefetching too early occupies device memory and prefetching too late fails to hide transfer latency. Inappropriate KV-cache eviction would lead to unnecessary recomputation. ANON addressed this challenge by combining static analysis hints and runtime observation to instruct retention, eviction and prefetching.

We implement ANON in Jac ByLLM (Mars et al., 2023; Dantamarayana et al., 2025) and use SGLang (Zheng et al., 2024) as the underlying model-serving engine. We evaluate ANON on four representative agent workloads imple-

mented in ByLLM Jac: a HoVer-style (Jiang et al., 2020) fact-checking agent, a HumanEval (Chen et al., 2021) coding agent, a BFCL (Patil et al., 2025) tool-use agent, and a FinanceBench (Islam et al., 2023) financial-analysis agent. We further evaluate a mixed workload comprising all four agent types to assess ANON’s end-to-end performance under concurrent, heterogeneous demand.

In summary, we make the following contributions: (1) We identify the inherent heterogeneity in programmatically generated prompts and its implications for serving. (2) We design and build ANON, a compiler-serving co-design that recovers and exploits prompt heterogeneity for efficient KV-cache management. (3) We conduct comprehensive evaluations of ANON against multiple established serving systems, demonstrating that ANON achieves up to 48% lower average JCT and 78% lower average TTFT than Vanilla SGLang, and  $1.46\times$  throughput on a mixed agent workload.

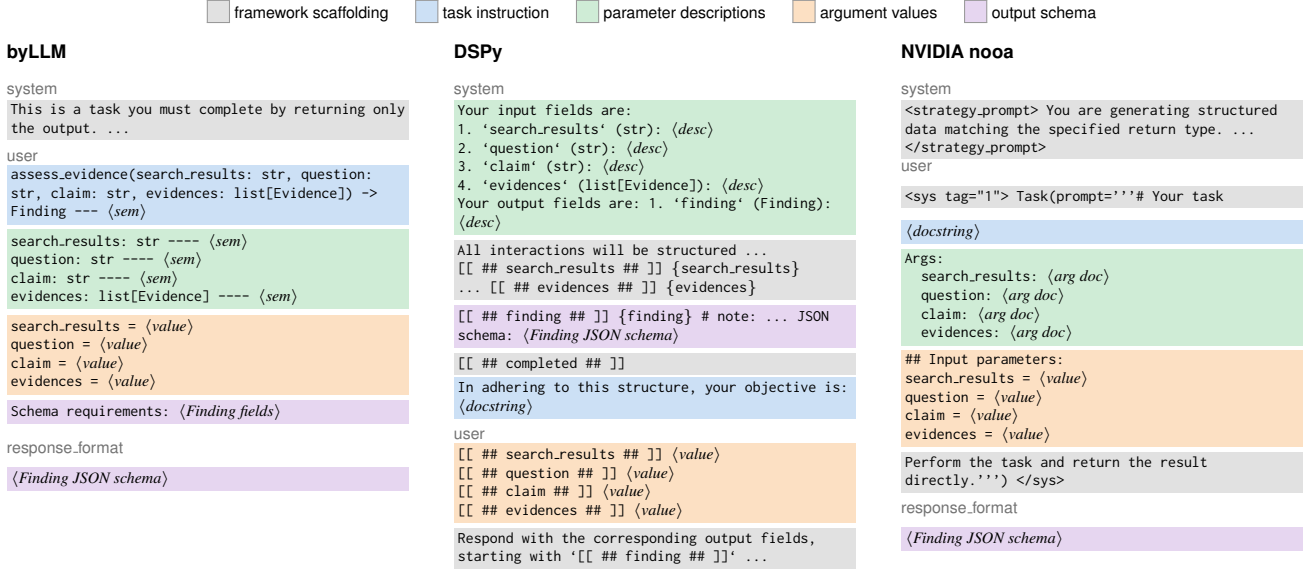
## 2 BACKGROUND

### 2.1 Serving systems for LLM agents

LLM serving engines retain the attention keys and values (KV) of tokens they have already processed to avoid repeating computation. PagedAttention manages these states in shareable memory blocks (Kwon et al., 2023), while SGLang indexes cached token sequences in a radix tree (Zheng et al., 2024). An incoming request reuses the longest cached prefix of its prompt and computes KV for the remaining suffix. Reuse stops at the first token mismatch: that token and every subsequent token must be re-computed, even if identical text appears later in a cached sequence, because its preceding context has changed. Prompt Cache supports modular reuse beyond a shared prefix, but requires modules to retain fixed, schema-assigned token positions (Gim et al., 2024).

Retained KV competes with running requests for device memory. Under memory pressure, engines such as SGLang evict unused cached prefixes by recency (Zheng et al., 2024). Hierarchical caching systems, including CachedAttention (Gao et al., 2024), Mooncake (Qin et al., 2025), and SGLang HiCache (SGLang Contributors, 2026), extend the cache beyond device memory with host memory and, optionally, storage. KV evicted from the device can remain in the host tier and be transferred back when needed; prefetching can overlap this transfer with other work. Reloading retained KV can be cheaper than recomputing it. The host tier is also limited: admitting one sequence consumes space that could retain another, so offloading does not eliminate cache-capacity pressure.

An agent *session* is one execution of an agent task, comprising a series of dependent LLM calls separated by tool and code execution. Between calls, the session’s KV sits



(*sem*), (*docstring*), (*desc*) and (*arg doc*) are the description strings each framework takes from the program (byLLM *sem* strings, DSPy Signature *docstring* and field *desc*, nooa function *docstring* and its *Args* entries); (*value*) is the runtime argument.

Figure 2. The prompt structure of LLM invocation function `assess_evidence`, lowered by three frameworks. They have similar structures: Same five kinds of component, only relative position differs.

idle and is exposed to recency eviction precisely while the agent is doing work that may lead to its reuse. Agent-aware serving systems use execution information to manage these gaps. KVFlow uses an agent step graph to estimate steps to future invocations and guide eviction and prefetching (Pan et al., 2025). Continuum retains KV across tool calls with a time-to-live (Li et al., 2025). CacheScout learns transitions between agents online to guide retention and prefetching (Zhang et al., 2026), while PBKV predicts future agent invocations from workflow history and the current context (Zheng et al., 2026). These systems share a request- or agent-step-level view of future execution: they organize cache management around the reuse of whole requests or agent steps.

## 2.2 Agent frameworks with LLM calls as programming abstractions

A class of agent frameworks exposes model-backed computation through typed programming interfaces. Examples include byLLM’s meaning-typed programming (MTP) abstraction (Dantanarayana et al., 2025), DSPy’s Signatures and Modules (Khatab et al., 2024), and NOAA’s typed Python methods (Furgale et al., 2026). SIGIL likewise compiles skill descriptions into harnesses containing typed model operations (Dantanarayana et al., 2026). The developer declares a function or signature describing the desired behavior, its inputs, and its return type; invoking it is an ordinary call within the program’s control flow. We use *call site* to refer these function invocation locations.

At each invocation, the framework runtime lowers the declaration and its argument values into model-facing messages. The declaration supplies the function name, *docstring* or semantic strings, parameter types and descriptions, and return schema. This information is known when the program is compiled: byLLM captures it in MTP’s meaning-based intermediate representation, MT-IR (Dantanarayana et al., 2025). The argument values become available only at runtime, when the call executes. Thus, a generated prompt combines a description fixed by the program with values supplied by that execution.

Some frameworks make deliberate choices about how to arrange this information. NOAA places static context first, appended event history next, and volatile state last to maximize KV-cache reuse across turns (Furgale et al., 2026). DSPy’s adapters use a fixed ordering of system instructions, demonstrations, conversation history, and current inputs; its ChatAdapter emits input fields in signature order (DSPy Contributors, 2026). These layouts follow hand-chosen conventions applied across call sites and programs, rather than an order derived from the properties of a particular program’s bindings. The framework knows the structure behind every prompt, but the serving engine receives only the token sequence.

## 3 MOTIVATION

**Runtime-lowered prompts have similar structures** Although these Program-Integrated LLM frameworks differ in surface syntax, they lower LLM invocations in similar ways.

They declare LLM invocations as functions or function-like declarations, and framework serialize that declaration together with the runtime arguments into a single prompt.

An example in Figure 2 demonstrate this. ByLLM in Jac, DSPy, NOAA lower the same `assess_evidence` LLM invocation. These prompts carry the same five kinds of components: framework scaffolding, task instruction, parameter descriptions, argument values, and output schema. The only difference is their relative position. What’s more, argument values are all emitted in declaration order.

**The component order carries no intent** The order in which these five components appear is decided solely by the framework runtime, without any program specific knowledge. Figure 2 showed the three representative frameworks arrange the same five components differently for the same equivalent invocation, which suggests that no particular order is required by the task. Even when argument values appear in declaration order, they do so simply because the runtime follows the parameter list, whose order was chosen by human developers.

**Prompt Components are heterogeneous** A prompt programmatically generated by an agent framework is not prose. Instead, it has inherent heterogeneity. The runtime constructs the prompt from different sources, such as program constants, runtime values, external inputs. Therefore these components have different sharing scopes, lifetimes and evolution pattern. As we discussed in §1 for Figure 1, this heterogeneity affects how the KV-cache can be managed effectively.

In summary, the heterogeneity within these programmatically generated prompt exposes new opportunities for efficient serving. However, none of these informations are served for effective serving, which motivates the design of ANON: Recovers heterogeneity by static analysis and use it to unlock more efficient KV-cache scheduling.

## 4 DESIGN

### 4.1 Overview

In this section, We present the design of ANON, an efficient serving engine for program-integrated LLM agents. ANON adopts a two-tier compiler-server co-design. On compiler side, ANON performs static analysis to gather heterogeneity metadata, and on the server side, ANON leverages this metadata to optimize KV-cache scheduling. We first introduce an interface *Agent Descriptor* to bridge the compiler and server, together with the methodology of their construction. Second, we present a heterogeneity-aware KV-cache scheduling strategy that leverages the agent descriptor to enhance serving performance.

### 4.2 Agent Descriptor

Here we define *Agent Descriptor* to bridge static analysis and server-side scheduling. An agent descriptor contains two parts: (i) **Call-site Template**: It provides a heuristic of the components’ lifetime, sharing scope and evolution pattern. (ii) **Workflow Automaton**: It contains the program-level structure of an agent, including workflow dependencies, execution time estimation and transition probabilities. ANON’s two-tier design let the compiler to provide a skeleton, while the server refines the weight with runtime observations. Two sides are therefore connected by this agent descriptor, and jointly serve a more efficient KV-cache scheduling strategy.

#### 4.2.1 Call-site Template

As discussed in §3, the components of a programmatically generated prompt can be classified into the following two categories: (i) **Invariant Specifications**: the components that are fixed once the program is compiled, such as the system prompt, response schema, and the tool declarations. They are rendered from the compiler’s own intermediate representation and stays constant. (ii) **Dynamic Bindings**: because LLM calls are declared as ordinary functions, the arguments of a call site are supplied in runtime. These runtime values are the dynamic bindings.

We therefore define the call-site template as

$$\mathcal{T}_c = \langle \mathcal{I}_c, \mathcal{B}_c, \mathcal{H}_c \rangle,$$

where  $\mathcal{I}_c$  is the invariant specifications,  $\mathcal{B}_c$  is the dynamic bindings, and  $\mathcal{H}_c$  is the heterogeneity annotation over those bindings. All three are recovered by static analysis.

**Heterogeneity annotation.** For each dynamic binding,  $\mathcal{H}_c$  records both how its value evolves from one call to the next and how long that value stays valid:

$$\mathcal{H}_c : \mathcal{B}_c \longrightarrow \mathcal{E} \times \mathcal{S}.$$

The evolution component  $\mathcal{E}$  classifies a binding by its value provenance. We distinguish  $\mathcal{E}$  into five classes, listed from the most to the least reusable: (i) **CONST**: the value is fixed by the program text and is identical at every call. (ii) **COPY**: the value is the same as the value another call site carries, so the bytes are already resident once that call site has run. (iii) **RESP**: the value comes from the reply produced by another call site. (iv) **EXTEND**: the binding is an append-only context, and the previous value remains a prefix of the new one. (v) **VOLATILE**: the value is freshly produced at each call, or its origin lies outside what the analysis can see. We write  $\rho$  for the rank of a class in the order above, which expresses how much of a binding’s bytes an earlier call can already supply.

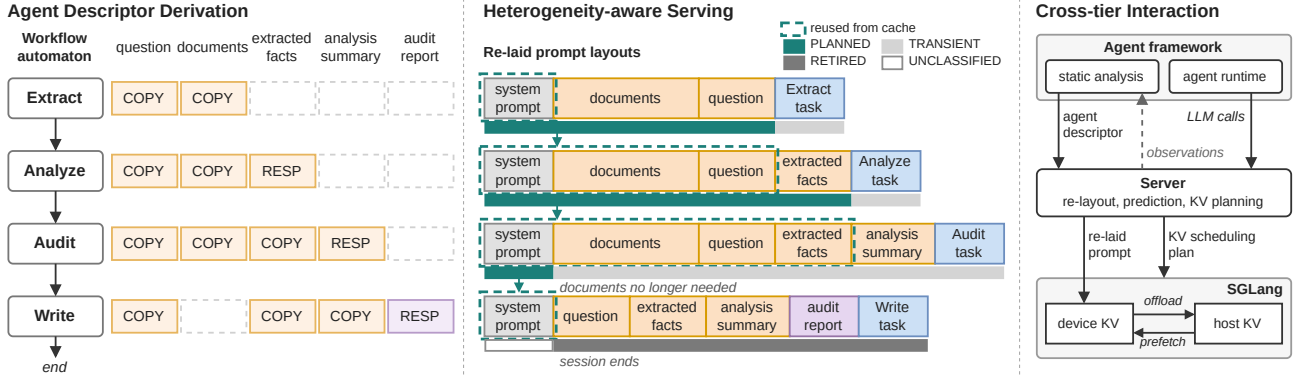


Figure 3. Architecture overview of ANON. Static Analysis derives agent descriptor, mark the intra-prompt heterogeneity. Server side then utilizes it for ANON’s efficient KV-cache scheduling.

The lifetime component  $\mathcal{S}$  names the program scope whose completion invalidates a binding’s value ( $\mathcal{S} = \emptyset$  meaning that the value survives for the whole session once produced). Lifetime is independent of evolution, and the server needs both. The evolution class governs which bytes a later prompt can reuse, while the lifetime governs when cached bytes can be given up.

#### 4.2.2 Workflow Automaton

An agent typically consists of a sequence of dependent LLM calls whose workflow structure is implicit in the program. Concretely, we abstract an agent as a time-annotated probabilistic finite-state machine (TPFM). We formulate the TPFM as:

$$\mathcal{A} = (Q, P, q_0, F, \mathcal{T}),$$

where:

1.  $Q$  is the set of states. Each state  $q \in Q$  is labelled by an LLM callsite;
2.  $P$  denotes the transition probabilities between states.
3.  $q_0 \in Q$  is the entrance callsite.
4.  $F$  specifies the termination probability of the agent at each state.
5.  $\mathcal{T} = \{D_q^{\text{exec}}, D_{q,q'}^{\text{gap}}\}$  denotes the timing annotations associated with states and transitions, where  $D_q^{\text{exec}}$  is the execution-time distribution of the LLM callsite represented by state  $q$ , and  $D_{q,q'}^{\text{gap}}$  is the inter-call gap distribution between the completion of callsite  $q$  and the invocation of its successor  $q'$ .

Static analysis fixes the shape of this automaton, while its weights comes from runtime logs. The states, the entry, the transitions among them, and final states are recovered from

code.  $\mathcal{T} = \{D_q^{\text{exec}}, D_{q,q'}^{\text{gap}}\}$  and  $P$  are often unknown at compile time and model dependent. The server observes these online and accumulates them across sessions, updating the statistical model of the agent consistently. Together the two halves give the server a forward-looking view of a session.

### 4.3 Adaptive Prompt Re-layout

Prefix reuse stops at the first byte that differs from a cached prompt. The message a framework emits for a call site follows the declaration order of the function, which may place a changing value ahead of values that later calls could reuse. ANON’s adaptive re-layout permutes the components of each call message so that the bytes most likely to be resident already come first. The descriptor  $\mathcal{T}_c$  (§4.2.1) and  $\mathcal{A}$  (§4.2.2) determine a partial order over the components, and the server completes it into a layout with the token lengths it observes at runtime.

#### 4.3.1 Movable Components

First, the dynamic bindings  $\mathcal{B}_c$  can be permuted freely. The compiler lowers each argument into a self-contained component labelled by its parameter name, and the framework places these components in declaration order only because that is the order of the argument list; Re-layout may therefore choose any order of the bindings.

Second, part of the invariant specification  $\mathcal{I}_c$  is movable as well.  $\mathcal{I}_c$  is a simplified abstraction. More precisely, it consists system prompt, reponse schema, tool schema and instruction specification (Task description and parameter description generated by compiler). Among them, system prompt stays universally constant, and two schemas are digested by model separately. Instruction header is movable.

ANON therefore operates on the order of bindings and the placement of instruction specification. Section 4.3.2 de-

scribes how ANON chooses a layout within this space.

#### 4.3.2 Layout Decision

Each call site sorts the bindings it carries by a program-wide key. Generally, *shared* bindings, which are carried by multiple call sites, are placed at the front. Then the bindings carried by a single call site. Algorithm 1 summarizes the procedure.

**Ordering private bindings.** The bindings carried by a single call site are sorted in ascending lexicographic order of

$$K(b) = (\text{scoped}, \rho, \text{depth}, -\text{length}),$$

whose components are defined as follows: **(i) *scoped***: the lifetime of  $b$ ; 0 if it is session-wide ( $S = \emptyset$ ) and 1 otherwise, so bindings that survive scope boundaries precede those reset by them. **(ii)  $\rho$** : the stability rank defined in §4.2.1, from 1 for CONST to 5 for VOLATILE, so the more reusable class leads. **(iii) *depth***: The number of transitions to the first call site that carries it from entrance, so values that enter a session earlier lead. **(iv) *length***: the token length of the value of  $b$ , observed by the server; negated so longer bindings precede.

**Ordering shared bindings.** The ordering of shared bindings is derived from program-level structures. For every transition from  $q$  to  $q'$ , Static Analysis identifies how each shared binding of the two call sites evolves. We categorize these patterns into three groups and ordered in following sequence: **(i)** Both call sites carry it and its value does not change between the two calls. **(ii)** Both call sites carry it and the program only appends to its value. **(iii)** Program rewrites it between the two calls, or only one of the two call sites carries it.

ANON collects these requirements over all transitions and orders the shared bindings to satisfy them; when several bindings can go next, the one with the smallest **(-length, depth)** is placed first. Bindings with conflicting requirements are clustered and sorted by the same key.

**Placing the specification.** The *instruction specification* is lowered from code by the compiler and is unique to each call site. Emitting it first would therefore prevent prefix sharing among different call sites, however many values they have in common. After sorting, ANON moves the specification behind the bindings whenever the leading binding is shared, so that sibling call sites can share large blocks of common bindings. A per-program switch keeps the specification in front when moving it is suspected of changing the model's behavior.

#### Algorithm 1 Layout decision algorithm

**Require:** Descriptor  $(\{\mathcal{T}_c\}_{c \in C}, \mathcal{A})$ , lengths  $\ell$ , switch *pin*

**Ensure:** Layouts  $\{L_c\}_{c \in C}$

```

1:  $J(b) \leftarrow (-\ell(b), \text{depth}(b))$ 
2:  $B_{\text{sh}} \leftarrow \{b : |\{c \in C : b \in \mathcal{B}_c\}| > 1\}$ 
3:  $R \leftarrow \emptyset$ 
4: for all  $e = (q, q') \in \text{EDGES}(\mathcal{A})$  do
5:    $B_e \leftarrow B_{\text{sh}} \cap (\mathcal{B}_q \cup \mathcal{B}_{q'})$ 
6:    $U_e \leftarrow \text{UNCHANGEDBINDINGS}(e, \mathcal{B}_q \cap \mathcal{B}_{q'})$ 
7:    $A_e \leftarrow \text{APPENDONLYBINDINGS}(e, \mathcal{B}_q \cap \mathcal{B}_{q'})$ 
8:   for all  $b \in B_e$  do
9:      $t_e(b) \leftarrow \begin{cases} 1 & b \in U_e \\ 2 & b \in A_e \\ 3 & \text{otherwise} \end{cases}$ 
10:  end for
11:   $R \leftarrow R \cup \{(b, b') \in B_e^2 : t_e(b) < t_e(b')\}$ 
12: end for
13:  $D \leftarrow \text{CLUSTERCONFLICTS}((B_{\text{sh}}, R))$ 
14:  $J_{\min}(S) \leftarrow \min_{b \in S} J(b)$ 
15:  $\sigma \leftarrow \text{TOPOSORT}(D, J_{\min})$  ▷ Min-key first
16:  $\pi_{\text{sh}} \leftarrow \langle \rangle$ 
17: for all  $S \in \sigma$  do
18:    $\pi_{\text{sh}} \leftarrow \pi_{\text{sh}} \parallel \text{SORT}(S, J)$ 
19: end for
20: for all  $c \in C$  do
21:    $\pi_c \leftarrow \langle b \in \pi_{\text{sh}} : b \in \mathcal{B}_c \rangle$ 
22:    $\pi_c \leftarrow \pi_c \parallel \text{SORT}(\mathcal{B}_c \setminus B_{\text{sh}}, K)$ 
23:    $s_c \leftarrow \text{INSTRUCTIONSPEC}(\mathcal{I}_c)$ 
24:   if  $\neg \text{pin} \wedge (\mathcal{B}_c \cap B_{\text{sh}} \neq \emptyset)$  then
25:      $L_c \leftarrow \pi_c \parallel \langle s_c \rangle$ 
26:   else
27:      $L_c \leftarrow \langle s_c \rangle \parallel \pi_c$ 
28:   end if
29: end for
30: return  $\{L_c\}_{c \in C}$ 

```

#### 4.4 Proactive KV-Cache Scheduling

While adaptive prompt re-layout improves prefix reuse only by reshaping the request, ANON's KV-cache scheduling proactively arrange the placement of KV cache entries. Using the agent descriptor, the server decides for every served KV-cache sequence what to keep, what to give up, and what to prefetch.

##### 4.4.1 Reusable Head and Volatile Tail

Since re-layout already places the stable and shared bindings at the front, a served prompt naturally divides into two segments: a reusable head and a volatile tail. In a full prompt, what behind the reusable head are all classified into volatile tail.

Reusable head consists two parts: a *static prefix* and a *session-shared extension*:

- **Static Prefix** is what the call site emits without run-time inputs, and its values outlives single session. It includes the invariant specifications  $\mathcal{I}_c$  together with the CONST bindings that re-layout has placed right behind them.
- **Session-shared Extension** is the run of bindings behind the static prefix that a later call of the same session will emit again at the same position.

The two parts are derived from agent program itself.  $\mathcal{H}_c$  names the provenance of each binding, and the automaton  $\mathcal{A}$  names the transitions between call sites. Additionally with the decided layout of each call-site, we determine the reusable head by jointly analyze them.

#### 4.4.2 Next-Call Prediction

Because the workflow automaton  $A$  already exposes all potential next calls, our prediction focus on the following three aspects: (i) The probability of the arrival. (ii) When will the call arrive. (iii) Which prefix of its prompt is can be speculatively constructed.

**Arrival probability.** We estimate the probability of the next call by a loop-tagged table. As the compiler side emits the workflow graph, ANON adopts Bourdoncle-style (Bourdoncle, 1993) weak topological ordering to discover all loops within the agent, so that each call-site  $q$  is assigned the chain of loops enclosing it. The server counts the iterations of each loop per session, so every call carries a tuple  $\zeta$  of the current iteration numbers of the loops enclosing its call-site, outermost first. We refine the transition and termination probabilities of  $\mathcal{A}$  by this tuple and define the table as

$$\mathcal{P}(q' \mid q, \zeta) \quad \mathcal{F}(q, \zeta)$$

where  $\zeta$  is the loop indices;  $\mathcal{P}$  stands for the probability that a call-site  $q$  in context  $\zeta$  is followed by a call-site  $q'$ ;  $\mathcal{F}$  stands for the probability that the session ends at call-site  $q$  under context  $\zeta$ . With an empty  $\zeta$  they reduce to  $P$  and  $F$  of §4.2.2.

The table is a frequency count that the server updates across sessions. When a call arrives, its call-site is counted as the outcome of the previous call-site under the previous call's  $\zeta$ , and termination is counted when the session closes. Each observation is also counted under every suffix of  $\zeta$  down to the empty tuple, so that for the same  $q$  the table holds one row per level of context, from the fully specified iteration to the pooled count over all iterations. After a call at  $q$  lands, we compute the probability of each outcome  $o$  by Witten–Bell interpolation from the empty context  $\zeta_0$  to

the full context  $\zeta_m = \zeta$ , where  $o$  ranges over  $\text{succ}(q)$ , the static successors of  $q$  plus a virtual outcome end for session termination:

$$\hat{P}_k(o) = \lambda_k \frac{c_k(o)}{n_k} + (1 - \lambda_k) \hat{P}_{k-1}(o),$$

$$\hat{P}_0(o) = \frac{1}{|\text{succ}(q)|}, \quad \lambda_k = \frac{n_k}{n_k + d_k},$$

where  $\zeta_k$  is the suffix of  $\zeta$  that keeps only the innermost  $k$  indices, and the three counts are read from the row  $(q, \zeta_k)$  of the table:  $c_k(o)$  is how many times outcome  $o$  followed a call at  $q$  in context  $\zeta_k$ ,  $n_k = \sum_o c_k(o)$  is the number of observations in that row, and  $d_k$  is the number of distinct outcomes with  $c_k(o) > 0$ .

Then,

$$\mathcal{P}(q' \mid q, \zeta) = \hat{P}_m(q') \quad \mathcal{F}(q, \zeta) = \hat{P}_m(\text{end})$$

A level with no observation is skipped. The weight  $\lambda_k$  grows with the support of a level, so a well-observed iteration dominates the prediction while a sparse one defers to the pooled behavior of  $q$ .

**Arrival time.** The time of arrival is conservatively approximated based on record statistics. The workflow automaton records  $D_{q,q'}^{\text{gap}}$  per transition: each time a call at  $q'$  arrives, the server appends the interval between the reply of the preceding call at  $q$  and this arrival. The interval is the agent's own time spent in its code between two calls.

We estimate the gap before the next call as:

$$\hat{g} = t_{50} - \frac{1}{2}(t_{90} - t_{50})$$

where  $t_{50}$  and  $t_{90}$  are the 50th and 90th percentiles of the gap samples on the transition  $(q, q')$ ; the predicted call is then expected  $\hat{g}$  after the reply of the current call. We introduce the second term to make sure a transition with more sparse distribution of gap receives a tighter estimation.

**Prefix reconstruction.** As indicated in §4.4.1, the prompt of the known successor  $q'$  is reconstructible, or at least partial-reconstructible. ANON builds it prompt in the layout  $L_{q'}$ , using the values that the session already carries. The construction processes bindings in the order of  $L_{q'}$  while consulting their evolution class, deducing value accordingly. The construction stops at the first binding with no known origin. Then the matched prefix is sent to kv-planner for speculative preparation.

#### 4.4.3 Lifecycle Bands

As we discussed in §3, different components have different lifetime, sharing scope and evolution patterns. Therefore, relying on recency based LRU would lead to suboptimal performance. ANON adopts a banded priority and steers KV-cache upon it.

**Bands.** We categorize KV-cache entries into four bands, from first evicted to last:

1. **RETIRED:** the volatile tail of prompt belonging to completed sessions.
2. **TRANSIENT:** the volatile tail of a served prompt of a live session. The session that produced them will not present them again.
3. **UNCLASSIFIED:** entries that analysis cannot categorize, such as the prompts of a program that was never registered. They keep the engine’s default treatment.
4. **PLANNED:** prefixes that a predicted call will reuse. Inside this band entries are ordered by the expected value of keeping them resident, defined next.

Eviction empties the lowest band first and falls back to recency only within a band, so KV-cache that will likely be reused left memory later. On the host tier, however, it keeps its recency order except that retired segments go first, so a head the device gave up under pressure remains reloadable while it may still be needed. Additionally, KV-cache entries marked as *TRANSIENT* won’t be admitted into host as they rival memory space with reusable KV-cache entries.

**Band transitions.** Bands are assigned at three moments of a session’s lifecycle. (1) **When a served prompt enters the cache**, ANON splits it at its reusable head (§4.4.1). The volatile tail and the reply generated after it become *TRANSIENT*; the head stays *UNCLASSIFIED* until the next prediction refresh finds a successor call that starts with it, which promotes it to *PLANNED*. (2) **When a later call arrives**, the transition it takes may rewrite bindings that earlier heads carried. Each affected head is cut back to the prefix that survives the rewrite, and the bytes past it are demoted to *TRANSIENT*. (3) **When the session ends**, the volatile tails of every prompt it served drop to *RETIRED*. Their heads stay *UNCLASSIFIED* and age out by recency, since other sessions of the same program may still share them.

These bands are mirrored directly into the radix tree’s eviction order. The band is the primary key of the eviction heap and recency breaks ties.

#### 4.4.4 Retention and Prefetch

Based on prediction, kv-planner of ANON conducts protection. The duty of it is to ensure required prefix resides on the device, or at least on host. Each prediction is summarized into a tuple of reconstructed prefix, arrival probability and estimated deadline:

$$\text{Prediction} = (\pi_{q'}, \mathcal{P}(q' | q, \zeta), t_{\text{reply}} + \hat{g})$$

where  $\pi_{q'}$  is the prefix reconstructed for  $q'$  and  $t_{\text{reply}}$  is the time the current call replied. Predictions are refreshed at every call, and are dropped when the predicted calls arrive or expires. Protection acts on this tuple in two ways. *Retention* keeps  $\pi_{q'}$  from leaving the device, and *prefetch* restores the part of  $\pi_{q'}$  that has already been demoted to the host.

**Retention.** The planner ranks the predicted reusable KV-cache entries by the value of keeping  $\pi_{q'}$  resident at every refresh:

$$\mathcal{D}(q') = \frac{\mathcal{P}(q' | q, \zeta) \cdot |\pi_{q'}|_{\text{cached}}}{\max(t_{\text{reply}} + \hat{g} - t_{\text{now}}, \tau_{\text{min}})}$$

where  $|\pi_{q'}|_{\text{cached}}$  is the number of tokens of  $\pi_{q'}$  cached (both on host and device) and  $\tau_{\text{min}}$  is a floor that keeps a call due now from dominating. ANON’s kv-planner sort all predictions by  $\mathcal{D}$  in descending order, and admits them into *PLANNED* until the band covers a fixed fraction of the device tier. Within the band, their  $\mathcal{D}$  add up on every radix cache node, making shared head outranks a private one.

**Prefetch.** When part of  $\pi_{q'}$  has been demoted to the host, the planner prefetches it back to the device ahead of the estimated arrive time. A prefetch release deadline is therefore defined as:

$$t_{\text{release}} = t_{\text{reply}} + \hat{g} - \frac{|\pi_{q'}|_{\text{host}}}{B_{\text{h2d}}} - \frac{|\pi_{q'}|_{\text{uncached}}}{B_{\text{prefill}}}$$

where  $B_{\text{h2d}}$  and  $B_{\text{prefill}}$  are the measured host-to-device copy rate and prefill rate. Among released predictions the earliest deadline goes first, and break ties by expected saving:

$$\text{Saving} = \mathcal{P}(q' | q, \zeta) \cdot |\pi_{q'}|_{\text{cached}}$$

A prefetch takes only free device space after reclaiming *RETIRED* and *TRANSIENT* entries, so it never interferes with real requests; when no space is free, prefetching pauses briefly but still prevent predicted KV-cache entries from being completely evicted. A prefetch is released when the engine is idle or its deadline  $t_{\text{release}}$  approaches.

## 5 EVALUATION

In this section, we evaluate ANON on multiple representative agent workloads, and ablate each components.

### 5.1 Environmental Settings

#### 5.1.1 Testbed and Model

We conduct all experiments on one NVIDIA RTX 3090 GPU using Qwen3-14B-AWQ. We allocate 5.2 GB of GPU memory to the device KV-cache pool and 10 GB of CPU memory to the host KV-cache pool.

### 5.1.2 Baselines

We compare ANON against the following serving systems, together with two ablated variants of ANON. We also run each established serving system with ANON’s adaptive prompt re-layout enabled: (i) **Vanilla SGLang** (Zheng et al., 2024) uses SGLang’s HiCache with LRU eviction. (ii) **CacheScout** (Zhang et al., 2026) identifies agents from their request prefixes and learns inter-agent execution transitions online. It uses the learned transition patterns to prioritize cached KV entries and prefetch the prefixes of predicted successor agents. (iii) **Continuum** (Li et al., 2025) retains each agent program’s KV cache for an adaptively selected time-to-live (TTL). The TTL is derived from observed inter-invocation delays and expected cache reuse, allowing the system to preserve reusable context while the agent executes tools. (iv) **KVFlow** (Pan et al., 2025) uses an Agent Step Graph to build a workflow-aware KV cache management framework for agentic workloads. (v) **ANON-KV-only** enables proactive KV-cache scheduling but disables adaptive prompt re-layout. (vi) **ANON-Relayout-only** enables adaptive prompt re-layout but replaces proactive KV-cache scheduling with LRU eviction.

### 5.1.3 Workloads

We evaluate ANON on four representative agent workloads, written with ByLLM (Dantanarayana et al., 2025) in Jac: a HoVer-style (Jiang et al., 2020) fact-checking agent, a HumanEval (Chen et al., 2021) coding agent, a BFCL (Patil et al., 2025) tool-use agent, and a FinanceBench (Islam et al., 2023) financial-analysis agent. We further evaluate a mixed workload comprising all four agent types to assess ANON’s end-to-end performance under concurrent, heterogeneous demand. The mixed workload runs 20 concurrent lanes, four per agent type; when a lane finishes a session, it immediately starts the next one.

## 5.2 End-to-End Performance of ANON

We tested ANON under different number concurrent workflows, and a mixed workloads. Across these workloads, we compare average job completion time(JCT), average time-to-first-token(TTFT) and throughput(prompt tokens served per minute) for the mixed workload.

**Performance on mixed workload** As shown in Table 1, ANON raises throughput to  $1.46\times$  that of Vanilla SGLang and reduces average TTFT by 25%. Continuum and CacheScout perform close to Vanilla SGLang on all three metrics. The gain of ANON comes from a higher cache hit rate in both device and host memory.

**Average JCT** Across the four workloads, ANON reduces average JCT by 6–48% compared with Vanilla SGLang (Fig-

Table 1. Performance on the mixed workload. Metrics are measured over each system’s steady-state window.

| System                                | Throughput   | TTFT        | Hit rate (%) |      |             |
|---------------------------------------|--------------|-------------|--------------|------|-------------|
|                                       | (ktok/min)   | (s)         | Device       | Host | Total       |
| <i>Original prompt order</i>          |              |             |              |      |             |
| Vanilla SGLang                        | 90.8         | 41.6        | 6.2          | 0.9  | 7.1         |
| CacheScout                            | 92.3         | 41.7        | 6.4          | 0.2  | 6.6         |
| Continuum                             | 90.6         | 42.1        | 4.4          | 1.0  | 5.4         |
| KVFlow                                | 92.6         | 41.2        | 7.5          | 0.6  | 8.1         |
| ANON-KV-only                          | 105.7        | 35.3        | 7.6          | 6.9  | 14.5        |
| <i>With adaptive prompt re-layout</i> |              |             |              |      |             |
| ANON-Relayout-only                    | 108.8        | 35.6        | 5.0          | 7.5  | 12.6        |
| CacheScout                            | 113.2        | 34.4        | 9.4          | 5.3  | 14.7        |
| Continuum                             | 110.8        | 34.8        | 4.8          | 7.1  | 11.9        |
| KVFlow                                | 116.0        | 34.1        | 10.8         | 10.1 | 20.9        |
| ANON w/o prefetch                     | 129.3        | 29.7        | 9.3          | 20.9 | 30.2        |
| ANON                                  | <b>132.3</b> | <b>29.0</b> | 9.1          | 22.6 | <b>31.6</b> |

ure 4), and by 34% on the mixed workload (Table 1). When concurrency is low, adaptive prompt re-layout accounts for nearly all of the gain. As concurrency grows, the gain of re-layout alone shrinks; proactive KV-cache scheduling carries the gain: it evicts volatile tails first and protects the reusable heads that predicted calls will need. On fact check, scheduling contributes from the lowest concurrency.

Except for an 8–13% reduction on BFCL at  $c = 4$ , Continuum and CacheScout perform close to Vanilla SGLang. Both exploit predictability between invocations but treat each prompt as an opaque token sequence. In the original prompt order, volatile bindings often precede reusable ones, making prefixes hold little that later calls can reuse. ANON-KV-only, which schedules on the original prompt order, shows the same limit.

**Average TTFT** Across the four workloads, ANON reduces average TTFT by 12–78% compared with Vanilla SGLang (Figure 4), and by 25% on the mixed workload (Table 1). As with JCT, adaptive prompt re-layout accounts for nearly all of the gain at low concurrency, and proactive KV-cache scheduling carries it as concurrency grows. On fact check and BFCL, and on finance at low concurrency, the TTFT reduction exceeds the JCT reduction, because a reused prefix skips its prefill, whereas JCT also includes decoding and tool execution which the KV-cache does not shorten. Continuum and CacheScout show the same limit as in JCT.

**Task Quality** We therefore compare task accuracy between the five systems that serve the original prompts and the that serve the re-laid prompts, using a paired permutation test over inputs (Table 2). On all four workloads the difference is not significant ( $p \geq 0.66$ ) and no consis-

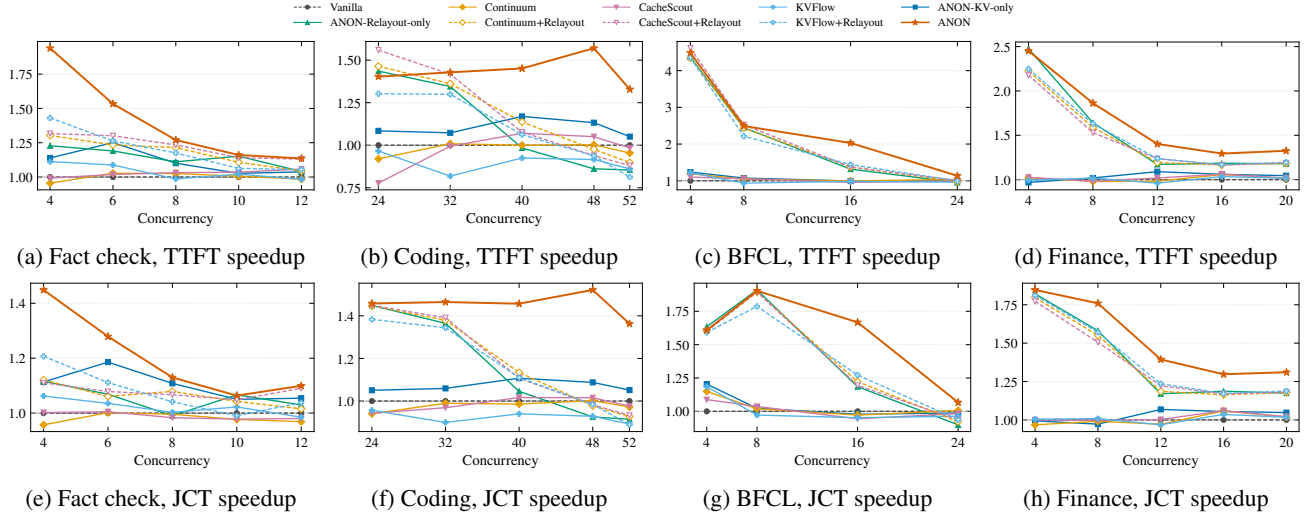


Figure 4. Average TTFT speedup (top) and average JCT speedup (bottom) over Vanilla SGLang under varying numbers of concurrent workflows.

Table 2. Task accuracy (%) of systems serving the original and the re-laid prompts, and  $p$ -values of a paired permutation test.

| Workload   | Original | Re-laid | $\Delta$ (pp) | $p$  |
|------------|----------|---------|---------------|------|
| Fact check | 20.0     | 20.4    | +0.4          | 0.95 |
| Coding     | 92.1     | 91.8    | -0.2          | 0.66 |
| BFCL       | 7.2      | 6.4     | -0.8          | 1.00 |
| Finance    | 34.6     | 35.4    | +0.8          | 0.82 |

tent direction of change. The performance gains of ANON therefore do not come at the cost of task quality.

### 5.3 Cache Efficiency of ANON

Figure 5 shows the cache hit rate of prompt tokens at the same concurrency levels as Figure 4. On coding and BFCL, adaptive prompt re-layout alone brings the hit rate at low concurrency to that of ANON: it places the stable and shared bindings at the front, where they form a reusable head that later calls hit. As concurrency grows, the hit rate of ANON-Relayout-only drops quickly, because LRU eviction does not distinguish reusable heads from volatile tails. With proactive KV-cache scheduling, ANON keeps a higher hit rate than ANON-Relayout-only as concurrency grows, and on coding its hit rate stays nearly flat. On fact check, ANON’s hit rate exceeds that of ANON-Relayout-only from the lowest concurrency. Continuum and CacheScout stay close to Vanilla SGLang at every concurrency. ANON-KV-only stays well below ANON, since under the original prompt order the reusable head it can protect is short. The mixed workload shows the same pattern (Table 1).

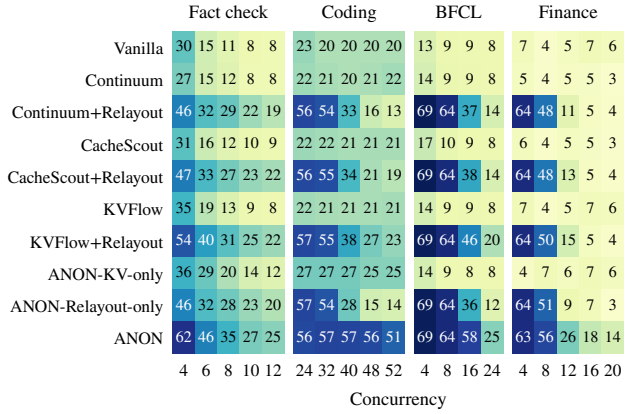


Figure 5. Cache hit rate (%) of prompt tokens under varying numbers of concurrent workflows.

### 5.4 Ablation Studies

We ablate ANON on the mixed workload by enabling its components one at a time (Table 1). ANON w/o prefetch keeps adaptive prompt re-layout and proactive KV-cache scheduling but disables prefetch. Figures 4 and 5 show ANON-KV-only and ANON-Relayout-only across concurrency levels.

Adaptive prompt re-layout alone raises throughput to  $1.20\times$  that of Vanilla SGLang and reduces average TTFT by 14%. Adding proactive KV-cache scheduling brings throughput to  $1.42\times$  and the TTFT reduction to 29%, and lifts the hit rate from 12.6% to 30.2%. Prefetch adds little on this workload ( $1.46\times$ , 30%), since under this load the device pool is rarely free enough for a prefetch to take place.

The two mechanisms depend on each other. Applied to the original prompt order, proactive KV-cache scheduling raises throughput by only 16% (ANON-KV-only); applied after re-layout, it raises throughput by 22% relative to ANON-Relayout-only. The combined 46% gain is larger than the sum of the two individual gains. Re-layout places the stable and shared bindings at the front, so a served prompt divides into a reusable head and a volatile tail, and scheduling gets opportunities to protect.

Besides, KV-cache scheduling and adaptive prompt re-layout are complementary. Applied on the re-laid prompts, CacheScout, Continuum and KVFlow all improve on ANON-Relayout-only (Table 1), and on the per-workload sweeps the gap widens with concurrency (Figure 4).

## 6 RELATED WORK

**KV-cache reuse** PagedAttention (Kwon et al., 2023) and SGLang (Zheng et al., 2024) reuse KV along an exact shared prefix, and hierarchical caches (Gao et al., 2024; Qin et al., 2025; SGLang Contributors, 2026) extend its capacity beyond device memory. Prompt Cache (Gim et al., 2024) and CacheBlend (Yao et al., 2025) reuse KV beyond the prefix, by fixing module positions in a schema or by selectively recomputing tokens of concatenated chunks. ANON keeps exact prefix matching and instead changes the prompt so that reusable components form the prefix; it is orthogonal to position-independent reuse.

**Existing agent serving systems** KVFlow (Pan et al., 2025), Continuum (Li et al., 2025), CacheScout (Zhang et al., 2026), PBKV (Zheng et al., 2026), and Pythia (Yu et al., 2026) predict future execution from step graphs, observed tool-call delays, learned transitions, or workflow histories, and use these predictions to retain and prefetch KV between calls.

**Existing Exploitation of prompt structure** Agent frameworks (Dantamarayana et al., 2025; Khattab et al., 2024; Furgale et al., 2026; Dantamarayana et al., 2026) know the structure of the prompts they generate. Where they arrange it for reuse, they apply one hand-chosen convention to every call site, such as NOAA’s static, history, and volatile regions. Parrot (Lin et al., 2024) passes application structure to the server through Semantic Variables that developers declare, but serves prompts in the layout the application wrote. SGLang’s frontend (Zheng et al., 2024) reorders prompt statements to lengthen the constant prefix, but has GPT-4 choose the order, which its authors note is not yet reliable.

## 7 CONCLUSION

The inherent structure and heterogeneity of programmatically generated prompts create opportunities for optimization. ANON identifies and exploits these opportunities rather than treating each prompt as a monolithic unit, thereby unlocking proactive optimizations that improve performance and efficiency. Our evaluation demonstrates that ANON’s methodology successfully creates a more efficient LLM serving system.

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